

# Abstract

Light availability in forest ecosystems is a key factor influencing various processes within the ecosystem. It affects tree growth performance, species composition, the microclimate, and soil formation processes. For a comprehensive understanding of forest ecosystems, information on local light availability and intensity is crucial to make informed decisions in research and forest management. Retrieving this information from measurements over large forest areas and at different points in time is a demanding task. However, for realistic representations, the need to consider the actual structure of a given stand is crucial for an accurate light simulation. In order to accurately simulate light interactions, we modeled a 3.5 ha area of natural mixed forest using LiDAR data obtained from unmanned aerial vehicles (ULS). In the next step, the data was classified to identify individual trees and assign additional semantic labels to each point, categorizing them into branches, leaves, and trunks. Subsequently, 3D mesh objects representing individual trees were generated using alpha shapes. To model the light distribution on the forest floor, the 3D objects are rendered in Blender using the Cycles render engine. By integrating the Sun Position add-on, Blender enables the simulation of different solar positions and the parameterization of sunlight impacting the scene. The final products are georeferenced maps from which the light intensity, represented as dimensionless pixel values, can be extracted for each pixel location. The data validation was conducted using a network of six pyranometers, whose measurements were compared with the corresponding pixel values. The Spearman correlation coefficient for two selected days in December averaged  $\rho = 0.87$  under diffuse conditions and  $\rho = 0.86$  under sunny conditions. However, analyses of daily light patterns reveal that on days with low solar angles, where no direct light penetrates the canopy gaps to reach the forest floor, the model fails to accurately capture expected light distributions. Conversely, at times with higher solar angles, where direct radiation reaches the forest floor, the model provides detailed representations of the light distribution. Validation of these observations under conditions of higher solar angles remains to be conducted. The developed method thus represents a low-threshold approach using ULS data to analyze direct sunlight exposure on the forest floor. However, it should be noted that the computation times are substantial, with model preparation requiring approximately 30 hours and simulating a single day's light patterns taking up to 18 hours. Future research could further improve the model, particularly regarding the calculation of radiative flux densities, in a radiative transfer model such as DART.